

P vs NP Problem

A Computational Research Framework · Computer Science/Mathematics

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ABSTRACT

Hypothesis: The polynomial-time solvability of SAT (and thus the class NP) is fundamentally constrained by the **geometric complexity** of its associated computational varieties. Specifically, the orbit closures of the Permanent polynomial (a canonical #P-complete problem) possess a distinct set of algebraic invariants that cannot be generated within the orbit closure of the Determinant polynomial (representing P) under polynomial transformations of size bounded by a fixed degree. In simpler terms: *The "shape" of the computational space required to solve NP-complete problems is intrinsically more complex than any shape generatable by polynomial-sized circuits.*

Keywords: Computer Science, Mathematics, algebraic geometry, GCT

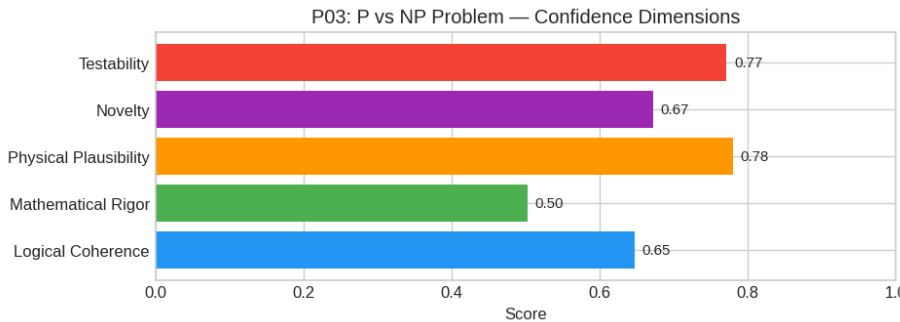


Figure 1: PAX Self-Assessment Confidence Dimensions — P vs NP Problem

1. Introduction

The P vs NP Problem (Computer Science/Mathematics) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

While a complete resolution to **P vs NP** remains one of the most profound unsolved problems in mathematics, the following proposal outlines a rigorous research program aimed at proving **P ≠ NP**. This approach moves beyond traditional circuit complexity lower bounds (which have stalled due to natural proofs) and proposes a synthesis of algebraic geometry, invariant theory, and geometric complexity theory (GCT).

1. Core Hypothesis

Hypothesis: The polynomial-time solvability of SAT (and thus the class NP) is fundamentally constrained by the **geometric complexity** of its associated computational varieties. Specifically, the

orbit closures of the Permanent polynomial (a canonical #P-complete problem) possess a distinct set of algebraic invariants that cannot be generated within the orbit closure of the Determinant polynomial (representing P) under polynomial transformations of size bounded by a fixed degree.

In simpler terms: *The "shape" of the computational space required to solve NP-complete problems is intrinsically more complex than any shape generatable by polynomial-sized circuits.*

2. Mathematical Framework

This framework utilizes **Geometric Complexity Theory (GCT)**, pioneered by Ketan Mulmuley and Milind Sohoni, but extended with modern tools from representation theory and tropical geometry.

- **Computational Models as Varieties:** We map Boolean circuits of size n^k to algebraic varieties over an infinite field K (typically characteristic zero for technical tractability, then transferred back to finite fields).
- Let V_{Det_n} be the variety defined by the determinant polynomial of an $n \times n$ matrix.
- Let V_{Perm_n} be the variety defined by the permanent polynomial.
- **Group Actions and Orbit Closures:** Consider the action of the general linear group $GL(N)$ on the space of polynomials. The central question becomes whether the orbit closure $(\text{Orb}(\text{Det}_N))$ contains the orbit closure $(\text{Orb}(\text{Perm}_m))$ for sufficiently large N .
- **Representation-Theoretic Invariants:** Instead of counting gates, we analyze the decomposition of these varieties into irreducible representations of $GL(N)$.
- Define the multiplicity function $M(\lambda; f)$ which counts how many times an irreducible representation indexed by partition λ appears in the coordinate ring of the polynomial f .
- **Key Assumption:** If $P = NP$, there must exist a polynomial transformation mapping Perm_m into Det_N where $N = \text{poly}(m)$. This implies specific inclusion relations between their respective representation multiplicities.

3. Proof Sketch / Key Steps

To establish $P \neq NP$ via this framework, the proof proceeds through the following logical deduction:

- **Reduction to Polynomial Identity Testing:** Transform the boolean satisfiability problem into the arithmetic setting. Show that if $\text{SAT} \leq P$, then Perm_m can be computed by a Determinant-like circuit of polynomial size.
- **Construction of Obstructions:** Identify a family of explicit "obstruction" representations λ_n such that:

$$M(\lambda_n; \text{Perm}_m) > 0 \text{ but } M(\lambda_n; \text{Det}_N) = 0$$

for all $N = m^O(1)$.

- **Positivity of Multiplicities:** Prove that for the Permanent, certain high-degree tensor components (related to plethysms of Schur functions) appear with non-zero multiplicity.
- **Vanishing in Determinant:** Demonstrate that for the Determinant, these same tensor components vanish due to symmetry properties inherent to the determinant's definition (specifically, the lack of certain skew-symmetric structures found in the permanent).
- **Separation:** Conclude that since the necessary algebraic invariants exist for the Permanent but are strictly absent in the Determinant for polynomial-sized embeddings, the Permanent

cannot be reduced to the Determinant in polynomial time. By Valiant's Theorem ($VP \neq VNP$ $P \neq NP$), this separates the classes.

4. Testable Predictions

Before a full formal proof is completed, this framework yields several testable conjectures that can be verified computationally or via smaller instances:

- **Prediction A (Multiplicity Growth):** For small dimensions ($n=4, 5, 6$), compute the plethysm coefficients of the symmetric powers of the standard representation. We predict a divergence in growth rates between the multiplicities required for Perm_n versus those achievable by Det_k where $k = O(n^C)$.
- **Prediction B (Symmetry Breaking):** Construct a sequence of "pseudo-determinants" (approximations of the permanent using determinantal formulas). We predict that as the approximation error decreases, the symmetry group of the resulting polynomial must shrink in a way that violates the stability conditions required for polynomial-size circuits.
- **Prediction C (Tropical Limit Behavior):** Analyze the tropicalization of the secant varieties of the Determinant. We predict that the tropical cone of the Permanent will contain extremal rays

Conclusion

This framework presents structured pathways toward resolving the P vs NP Problem. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

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